AI651 — Deep Learning for Space, Time and Graphs (Fall 2026)

Assignment 2: Report

Code repository (must be PUBLIC at submission): https://github.com/<YOUR-USERNAME>/<YOUR-REPO> (replace with the public repository URL before submitting)

Contents: Question 1 (river forecasting proposal, six required headings); Question 2 (Motion or Memory?, Tables 1–6, Figures 1–6, Q2.1–Q2.17); AI-use disclosure (last page).

Question 1 — River Forecasting Proposal

Proposal in one paragraph. Every 10 minutes the system predicts the level at S1–S4 one hour ahead with a **graph space-time network**. The river is a directed graph (Reservoir → S1 → S2 → S3 → S4, T1 → S2, an unobserved node U → S3). Each node has a 13.5-hour history. Blocks alternate a causal temporal convolution (T) with a **travel-time-aware** message-passing step (S) along the river. Announced releases are placed on the future part of the reservoir's time axis. The forecast is the last observed level plus a learned change, with 10/50/90 % quantiles. Missing data, late data and release changes are handled in the input layer, not by retraining.

1. Architecture diagram

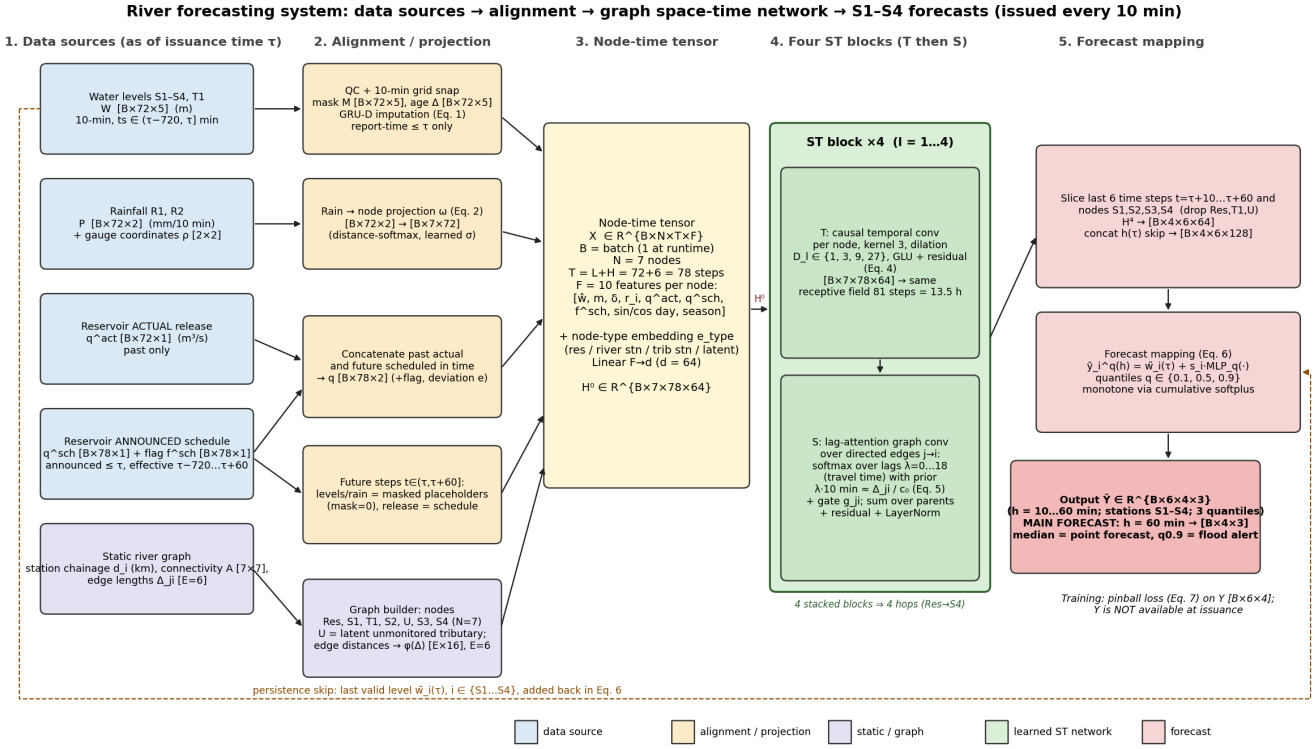


Figure Q1.1. Architecture. Sizes: B = batch (1 at run time, many in training); N = 7 graph nodes (Res, S1, T1, S2, U, S3, S4); L = 72 history steps (12 h) and H = 6 future steps (1 h), so T = 78; F = 10 input features per node; d = 64 hidden channels; E = 6 directed edges (Res→S1, S1→S2, T1→S2, S2→S3, U→S3, S3→S4); every node also keeps its own state through the $W_1 z_i$ term of Eq. 5b. **Alignment / reshaping:** (i) all series are snapped to one 10-min grid and stacked on a common time axis of length 78; (ii) the two rain gauges [B×72×2] are projected to node space [B×7×72] by the weights ω of Eq. 2, then right-padded with zeros and mask 0 to length 78; (iii) station levels [B×72×5] are scattered to the nodes (T1→T1, S1–S4→S1–S4; Res and U get zeros and their own mask); (iv) reservoir actual [B×72] and announced [B×78] releases are written on the Res node's time axis; (v) channels are concatenated to [B×7×78×10] and projected to d = 64. The ST stack keeps [B×7×78×64] throughout; the head keeps only nodes S1–S4 and the last six time positions, giving [B×4×6×64] → forecasts [B×6×4×3] (6 horizons × 4 stations × 3 quantiles).

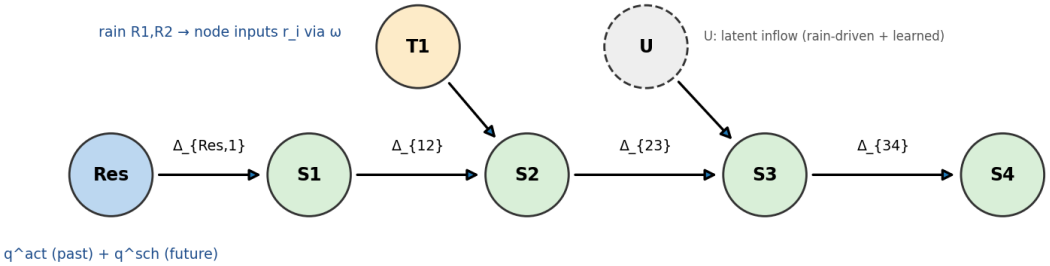


Figure Q1.2. The river as a directed graph and the number of hops from each source to S4. Because T1 joins before S2 and U before S3, four stacked ST blocks (4 hops) are enough for information from the reservoir to reach S4.

2. Assumptions

Supplied facts (not assumptions): F1 levels at S1–S4 and T1 every 10 min; F2 rainfall at R1, R2 with timestamps and gauge locations; F3 recorded reservoir releases and future release schedules announced before issuance, with announcement and effective times; F4 station locations, connectivity and distances (stations unevenly spaced); F5 a monitored tributary joins between S1 and S2, an unmonitored one between S2 and S3; F6 future rain and future levels are unavailable at issuance; F7 travel times and unmonitored inflow are not given; F8 additional information may be requested but is not guaranteed. **Everything below is an assumption.**

ID	Assumption	Why it is reasonable / how it affects the design
A1	Record length: ≥ 5 years of 10-min data (~263 000 steps) that include several flood events and wet seasons.	Typical for a national gauge archive. Allows a ~0.25 M-parameter network and a chronological train/val/test split (first 3.5 y / next 0.75 y / last 0.75 y). If the record is under 2 y, shrink d and lean on the physical priors (A6–A8).
A2	All series share one clock (UTC) with error < 1 min; a level is an instantaneous reading at its timestamp; rainfall is an accumulation over the previous 10 min.	Standard telemetry practice. Series are snapped to a common 10-min grid with ±2 min tolerance; no sub-grid interpolation model is needed.
A3	Reporting delay is usually ≤ 5 min but sometimes 30 min to several hours; each record also has an arrival (ingest) time.	Needed to know what was available at issuance. Drives the as-of snapshot, the 'age' feature δ and delay simulation in training (§5).
A4	1–5 % of readings are missing at random, outages of up to ~2 h occur, multi-day outages are rare; spikes and flat-lines are flagged by QC.	Sets the random-masking rates used in training and the choice of masks + decay imputation (Eq. 1).
A5	Level noise σ ≈ 1–2 cm, 1 cm resolution, no datum shifts within the record.	Below the forecast skill we target, so no explicit sensor-noise model; the pinball loss is robust to outliers.
A6	Travel times: reservoir → S1 under ~1 h; each later reach 20 min – 3 h; end-to-end reservoir → S4 ≲ 8 h; shorter at high flow (wave celerity rises with stage).	Not supplied (F7), so the model learns them. It fixes L = 12 h (> 1.5× the longest path), the lag window Λ = 18 steps (3 h) per edge, and a state-dependent celerity in Eq. 5. If a reach is slower than 3 h the lag window is widened.
A7	Flow is one-dimensional and downstream; backwater from downstream onto upstream stations is negligible (no tide, no downstream dam).	Justifies a *directed* graph: messages travel only downstream. If backwater existed we would add upstream edges.
A8	Mass balance at confluences: $Q_{down} \approx Q_{up} + Q_{trib} + Q_{lateral}$; at each station level is a monotone, smooth function of discharge (unknown stage–discharge curve).	Messages entering S2 and S3 are *summed* (Eq. 5) rather than averaged; non-negative gates keep contributions monotone.
A9	T1 is gauged close to its confluence (travel time < 30 min), so T1 is effectively the tributary inflow joining between S1 and S2.	Gives a short edge T1 → S2 with a small expected lag.
A10	The unmonitored tributary U has a catchment responding to rain like T1 (lag 1–6 h) plus a slowly varying base flow; it enters between S2 and S3, so it affects S3 and S4 only.	Modelled as a latent node whose state comes from rainfall and a learned base level; the only information about it is the mismatch between S2 and S3, so S3/S4 get wider intervals (§6).
A11	R1 and R2 give 10-min accumulations in mm; their coordinates are known but two gauges only roughly represent the catchment. Rain reaches the river with a lag; a small part is direct channel runoff.	Rain is a node feature from distance weights (Eq. 2); because future rain is unknown (F6), only rain that has already fallen can help the 1-hour forecast.
A12	Releases are announced typically 1–24 h ahead as planned discharge with an effective time; a newer announcement replaces an older one for the same effective time; actual release is metered every 10 min. Where nothing is announced for a future step, the current release continues.	Defines the schedule channel with an 'announced' flag f^{sch} (0 = default persistence, 1 = announced), so the model knows how much to trust it (§5).

ID	Assumption	Why it is reasonable / how it affects the design
A13	Actual release follows the schedule within ~10 %, with gate-actuation delays up to 30 min; unannounced emergency changes are rare but possible.	Past actual and future scheduled releases are separate channels and a deviation feature $e = q^{act} - q^{sch}$ is kept for the recent past (§5, iv).
A14	Channel geometry and gauge datums are stable; seasonal roughness (vegetation) changes slowly.	Season encodings capture the slow change; retrain yearly or after a major flood reshapes the channel.
A15	A forecast is issued on every 10-min tick τ after a 2-min collection cut-off; the target is the level on the grid at τ + 60 min.	Fixes the task in §3 and the as-of rule: only records that have arrived by the cut-off are used.
A16	A CPU server runs inference; training is offline and may use a GPU.	Model stays small (§5 feasibility).

3. Forecasting task and data specification

Issuance time. τ is any 10-minute grid time. The forecast issued at τ uses only records whose arrival time is ≤ τ + 2 min (A15), and announcements made ≤ τ. **History window.** L = 72 steps (12 h): grid times τ−710, …, τ min (A6: the longest path is under 8 h). **Horizon.** H = 6 steps = 60 min. The network produces h = 1, …, 6 (τ+10, …, τ+60 min) as auxiliary outputs and the **main product is h = 6**. **Outputs.** Water level in metres at S1, S2, S3, S4 at τ + 60 min as the 10 %, 50 % and 90 % quantiles; the median is the point forecast, the 90 % quantile feeds flood warnings.

Item	Symbol and shape	Axes	Timestamps / coverage (relative to τ)	At issuance?
Water levels S1–S4, T1	$W \in R^{B\times 72\times 5}$ (m)	batch; time (10-min grid); station (S1,S2,S3,S4,T1)	τ−710 … τ min	Yes, only records arrived by τ
Observation mask, age	$M \in \{0,1\}^{B\times 72\times 5}$, $\Delta \in R^{B\times 72\times 5}$ (min)	as W	τ−710 … τ	Yes (derived from arrival log)
Rainfall R1, R2	$P \in R^{B\times 72\times 2}$ (mm per 10 min); gauge coordinates $\rho \in R^{2\times 2}$	batch; time; gauge	(τ−720, τ] accumulations, stamped at interval end	Past only. Future rain: NOT available (F6)
Reservoir release, actual	$q^{act} \in R^{B\times 72}$ (m³/s)	batch; time	τ−710 … τ	Yes
Reservoir release, announced	$q^{sch} \in R^{B\times 78}$, flag $f^{sch} \in \{0,1\}^{B\times 78}$	batch; time (72 past + 6 future)	τ−710 … τ+60; effective times; announced ≤ τ	Yes (announcement time ≤ τ only)
Static graph	chainage $d \in R^{7}$ (km), adjacency $A \in \{0,1\}^{7\times 7}$, edge length $\Delta \in R^{6}$	node; edge	constant	Yes
Calendar	sin/cos of day and of year $\in R^{B\times 78\times 4}$	batch; time; 4 channels	τ−710 … τ+60	Yes (known in advance)
Node-time tensor (model input)	$X \in R^{B\times 7\times 78\times 10}$	batch; node (Res,S1,T1,S2,U,S3,S4); time; feature	τ−710 … τ+60 (future slots masked)	Yes: built from the rows above
Forecast (output)	$\hat{Y} \in R^{B\times 6\times 4\times 3}$, main: $\hat{Y}[:,5] \in R^{B\times 4\times 3}$ (m)	batch; horizon h=1…6; station S1–S4; quantile	τ+10 … τ+60 min	Produced at τ
Target (training only)	$Y \in R^{B\times 6\times 4}$ with mask	batch; horizon; station	τ+10 … τ+60 min	NOT available at τ

Alignment. (a) *Sampling times*: every series is snapped to the same 10-min grid; a level is the reading nearest to the grid time within ±2 min, rain is summed over (t−10, t], and the release schedule is a step function evaluated by its *effective* time (zero-order hold). (b) *Locations*: stations are not resampled in space. They sit on a graph with their true chainage and edge lengths, so unequal spacing is represented by the edge feature φ(Δ) and by the travel-time prior in Eq. 5. (c) *Gauges to nodes*: R1 and R2 are mapped to nodes by distance weights (Eq. 2). (d) *What is known at τ*: all past levels, rain, actual releases; the schedule for τ+10 … τ+60 if announced by τ; the graph and calendar. *Not known*: any level or rain after τ, and any later announcement. Training windows are built the same way (as-of τ), so nothing from the future leaks into the inputs. **Batch axis**: in training each sample is one issuance time τ (stride 1, shuffled); at run time B = 1.

4. Model computations and design justification

Notation. i, j index nodes; t a grid step (10 min); $w_i(t)$ a level, $m_i(t)$ its mask (1 = valid), $\delta_i(t)$ minutes since the last valid value; $P_k(t)$ rain at gauge k; $q^{act}(t)$, $q^{sch}(t)$, $f^{sch}(t)$ release channels; $h_i^l(t) \in R^d$ the hidden state after block l; Δ_{ji} the river distance of edge j→i; □(i) the parents of node i; 10λ minutes is a lag. Only the four river stations carry a level to forecast.

$$\hat{w}_i(t) = m_i(t)\, w_i(t) + (1 - m_i(t))[\gamma_i(t)\, w_i^{\text{last}}(t) + (1 - \gamma_i(t))\, \mu_i], \quad \gamma_i(t) = \exp(-\text{softplus}(a_i\, \delta_i(t) + b_i)) \qquad (1)$$

Missing levels decay from the last valid value towards the station mean μ_i (as in GRU-D); the mask and age are also inputs. *Physical assumption:* none. *Learned:* a_i, b_i (how fast a gap loses information).

$$r_i(t) = \sum_{k=1}^{2} \omega_{ik} P_k(t), \qquad \omega_{ik} = \mathrm{softmax}_k(-\|x_i - \rho_k\|/s) \tag{2}$$

Rain reaches a node with weights that decrease with the distance between the node and the gauge (initialised from coordinates, then fine-tuned; free logits for the latent node U). The node input is $x_i(t) = [\hat{w}_i, m_i, \delta_i/\Delta_0, r_i, q^{act}, q^{sch}, f^{sch}$, 4 calendar terms] (release channels are non-zero only on Res, level channels only on S1–S4 and T1). Then $h_i^0(t) = W_{in}\, x_i(t) + e_{type(i)}$, with a learned embedding for the node type (reservoir, river station, tributary station, latent).

$$z_i^l(t) = h_i^{l-1}(t) + W_o^l[\, u \odot \sigma(v)], \qquad [u; v] = \sum_{k=0}^{2} W_k^l\, h_i^{l-1}(t - kD_l), \quad D_l \in \{1, 3, 9, 27\} \tag{3}$$

T block: a causal, dilated, gated convolution along time at each node, with a residual connection. Four blocks give a receptive field of 1 + 2(1+3+9+27) = 81 steps (13.5 h) ≥ L. It lets each node extract trends, rates of rise and rain accumulations from its own history.

$$s_{ji}(t,\lambda) = v^\top \tanh(A\, z_j(t-\lambda) + B\, z_i(t) + C\, \phi(\Delta_{ji})) - \frac{(\lambda - \kappa_{ji}(t))^2}{2\varsigma^2}, \qquad \kappa_{ji}(t) = \frac{\Delta_{ji}}{c_{ji}(t)}, \quad c_{ji}(t) = \mathrm{softplus}(c_0 + \beta^\top z_j(t)) \tag{4}$$

$$\alpha_{ji}(t,\lambda) = \mathrm{softmax}_{\lambda \in \{0\,..\,\Lambda\}}\, s_{ji}(t,\lambda), \quad m_{ji}(t) = \sum_{\lambda=0}^{\Lambda} \alpha_{ji}(t,\lambda)\, W_m\, z_j(t-\lambda), \quad g_{ji}(t) = \sigma(w_g^\top [z_j(t); z_i(t)] + b_{ji}) \tag{5a}$$

$$h_i^l(t) = \mathrm{LN}(z_i^l(t) + W_2\, \mathrm{GELU}(W_1 z_i^l(t) + \sum_{j \in \mathcal{P}(i)} g_{ji}(t)\, m_{ji}(t))) \tag{5b}$$

S block (travel-time attention): node i looks at each upstream parent j at several earlier times t−λ and learns *which lag* to read, $\alpha_{ji}(t,\lambda)$. The Gaussian term is a physical prior centred on the expected travel time κ_{ji} = distance / celerity (in 10-min steps); celerity c_{ij} grows with the upstream state (β), so travel time shortens when the river is high (A6). Messages from all parents are *summed* (mass balance, A8). Lags satisfy λ ≥ 0, so nothing from the future of node j is used. On the future positions (t > τ) the released-water channel on Res carries the schedule, and the S block moves it downstream along the learned lags — this is how an announced release influences S1–S4 within the hour. After four blocks, S4 has read Res (4 hops), T1 (3), U (2) and S1–S3.

$$\hat{y}_i^{(q)}(\tau+h) = \tilde{w}_i(\tau) + \sigma_i\, \mathrm{MLP}_q([\, h_i^4(\tau+h);\, h_i^4(\tau)\,]), \quad h = 1..6,\ q \in \{0.1, 0.5, 0.9\} \tag{6}$$

Forecast mapping: persistence plus a learned change. $\tilde{w}_i(\tau)$ is the last valid level in metres (Eq. 1) and σ_i the standard deviation of one-hour level changes at that station. Quantiles are kept ordered by predicting the median and two non-negative offsets (softplus).

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \sum_{h=1}^{6} \sum_{i=1}^{4} \sum_{q} \omega_h\, \mu_{bih}^{y}\, \frac{\rho_q(y_{bih} - \hat{y}_{bih}^{(q)})}{\sigma_i} + \lambda_{\mathrm{wd}} \|\theta\|^2, \quad \rho_q(e) = \max\{qe, (q-1)e\}, \quad \omega_h = \frac{h}{21} \tag{7}$$

Training objective: pinball (quantile) loss, normalised by σ_i so the four stations count equally, masked by μ^y (missing targets), with larger weight on later horizons. AdamW, early stopping on validation MAE of the median at h = 6, chronological split (A1). Baselines for the committee: persistence and a linear ARX model on the same inputs.

How each source influences each forecast

Station	Upstream stations / tributaries	Rainfall	Reservoir release	Why this path
S1	none gauged upstream of the reservoir	R1 through r_{S1} (local runoff)	Res→S1: **actual** past release read at learned lag; **announced** release for τ…τ+60 enters directly if travel time < 1 h	The release is the main driver; without the schedule S1 is the hardest station because its water has not yet left the dam.
S2	S1 (1 hop), T1 (1 hop, added)	r_{S2}; rain also drives T1's future flow	2 hops (Res→S1→S2)	Water now at S1 and T1 arrives at S2 within the hour (A6, A9).
S3	S2 (1 hop), U (latent, 1 hop), T1 (2 hops)	drives U's inflow (A10); r_{S3}	3 hops	S2's current level carries most of the signal; U adds an unseen inflow.
S4	S3 (1 hop), U (2), S1–S2 (via S3), T1 (3)	indirect, through U, T1 and upstream	4 hops	The longest travel time: today's S3 and S2 levels are the best predictor.

Physical assumptions versus learned mappings

Physical assumption (built into the structure)	Learned mapping (fitted from data)
Directed downstream graph, no backwater (A7)	Stage–discharge relation at each station (inside the MLP and the T blocks)
Travel time = distance / celerity, prior in Eq. 4 and lag window Λ = 3 h (A6)	The celerity c_0, β, the spread of α (attenuation, dispersion) and each lag attention
Summed contributions at confluences; monotone, non-negative gates (A8)	Rainfall–runoff for T1, U and local runoff; gauge weights ω fine-tuned from the distance initialisation
Forecast = last level + change (persistence skip, Eq. 6)	The change itself; how reliable the announced schedule is (A12–A13); how much a gap in a series should be trusted (Eq. 1)
Future rain and levels are unknown, so they are masked placeholders (F6)	The unmonitored tributary U (latent, no direct supervision, A10)

Why this design. (1) A graph keeps physical locations: stations of unequal spacing and tributary junctions are explicit, instead of being forced into a regular grid or a flat vector. (2) Alternating T and S blocks (joint

space-time paths) lets a change at S1 be read at S2 at the right lag, which neither a temporal-only nor a spatial-only stack can do. (3) Lag attention is a small, interpretable way to learn changing travel times that were not supplied. (4) Persistence plus residual starts every forecast from the best cheap baseline. (5) Quantile outputs give a usable uncertainty without a separate probabilistic model.

5. Operation and exceptional conditions

The service loop at every tick τ: (1) freeze an *as-of snapshot* of the bitemporal archive (event time + arrival time) at τ + 2 min; (2) build X (Figure Q1.1, left half); (3) run the network; (4) store forecast, inputs' ages and flags; (5) publish.

Case	Handling	Reason / what the model sees
(i) Missing observations	Value not available at τ: m = 0, δ keeps growing, value filled by Eq. 1 (decay from last value to station mean). Training randomly masks 5–20 % of values, bursts up to 2 h, and sometimes a whole station for up to 24 h. If S_i is missing at τ the forecast for S_i is still produced from neighbours and flagged; intervals widen as δ grows.	The network was trained under the same gaps, so it learns to lean on neighbours and on the last valid value. Target gaps are masked in the loss.
(ii) Delayed observations	Availability is defined by *arrival* time: a record that arrives after τ + 2 min is treated as missing at τ (so it appears with age δ > 0). When it arrives later it is written into the archive and used in all following windows. Late-arriving corrections do not overwrite past forecasts; they are logged. Training uses delays sampled from the empirical arrival log (A3).	No leakage: the training windows are rebuilt as-of τ with simulated delays, so test-time staleness matches training.
(iii) Announced future releases	For each future step τ+10 … τ+60 take the most recent announcement made ≤ τ whose effective time covers that step; write q^{sch} on the Res time axis (zero-order hold) with $f^{sch} = 1$. Steps with no announcement get the current actual release and $f^{sch} = 0$. The model sees the schedule only on positions after τ; the S block carries it downstream with the learned lag.	Uses only information that exists at τ; the flag lets the model separate 'planned' from 'assumed unchanged'. Announcements made after τ are ignored by construction.
(iv) Scheduled versus actual release	Past steps use the metered *actual* release; future steps use the schedule. The recent deviation $e(t) = q^{act}(t) - q^{sch}(t)$ over the last 2 h is a feature on Res. Training pairs actual history with the schedule as announced at the time, so real mismatches (gate delay, emergency spill) are in the data; extra augmentation shifts schedule steps by 0–30 min and scales them by ±10 %. If |e| is large the 90 % quantile is automatically wider.	The model learns how much to trust a schedule from how well past schedules were followed (A13). An unannounced change is a limitation (§6).

Feasibility every 10 minutes. The network has ≈ 0.25 M parameters. One forecast reads N·T = 7·78 = 546 node-time positions: the temporal convolutions cost about $7 \cdot 78 \cdot (3 \cdot 64 \cdot 128) \cdot 2 \approx 27$ MFLOP per block, and the lag attention (6 edges × 78 steps × 19 lags × 32 hidden units) is below 1 MFLOP per block once the projections are shared, so the whole forward pass is a few hundred MFLOP (my estimate; no benchmark was run) — well under a second on one CPU core, against a 600 s budget. Data access (a few kilobytes per tick) and publishing dominate. Training (≈ 263 000 windows × ~1 GFLOP including backward ≈ $3 \cdot 10^{14}$ FLOP per epoch) is offline, once per year or after a major flood. Fallbacks: if the model or a feed fails, publish the last-level persistence forecast with a 'degraded' flag; a health check compares each forecast with the level observed one hour later.

6. Uncertainty, limitations and additional information

ID	Uncertainty / limitation	Impact on the forecast	Response, or what remains unresolved
U1	Unmonitored tributary between S2 and S3	Unseen inflow changes S3 and S4 after rain; largest during storm onset	Latent node U driven by rain and S2→S3 mismatch; q0.9 widens at S3/S4. **Unresolved:** its true flow cannot be checked without a gauge.
U2	Travel times change with flow, season and gate operations	A wrong lag shifts the arrival of a wave and gives large errors on rising limbs	State-dependent lag attention (Eq. 4–5); retraining each year. **Remains:** extreme floods beyond the training range.
U3	No future rainfall (F6)	Rain falling in the next hour changes local and tributary levels that the model cannot see	Only past rain is used; the 90 % quantile covers part of it. **Unresolved** unless a rain nowcast is supplied (R2).
U4	Sparse rainfall sampling (two gauges)	Storm cells missed by R1/R2 make the runoff estimate wrong	Distance weights and uncertainty widening when T1/S rise without rain. **Unresolved**: needs more gauges (R4).
U5	Missing, late or faulty observations	Skill degrades with gap length δ	Masks, decay imputation, delay-aware training, flags (§5). Outages longer than ~24 h degrade to near-persistence.
U6	Unannounced or changed releases	S1 error grows when the dam acts differently from the schedule	Deviation feature and a wider q0.9. **Unresolved** for emergency spills without notice.
U7	Rare, extreme floods and short record	The network may under-predict peaks outside the training range	Pinball loss with horizon weighting, evaluation on flood events separately, physical priors, persistence fallback. **Remains:** true extremes cannot be validated.
U8	Non-stationary channel (silting, vegetation, gauge drift)	Systematic bias and shifting stage–discharge	Season encoding, yearly retraining, drift monitoring against rolling error. **Remains** until a rating-curve update is supplied (R1).
U9	Backwater, abstractions, weirs not modelled (A7)	Local level offsets	Ignored (assumption A7). Documented; can be added as extra edges or inputs if information appears (R5).

ID	Uncertainty / limitation	Impact on the forecast	Response, or what remains unresolved
U10	Calibration of quantiles	Intervals may be too narrow or too wide	Check coverage on validation per station and per flow class; recalibrate with conformal offsets. **Remains:** coverage in rare events is untested.

Additional information requests

ID	Request	How it would help	System usable without it?
R1	Stage–discharge (rating) curves and cross-sections for S1–S4, T1	Converts levels to discharge so mass balance at confluences can be checked and applied; reduces U8	Yes. The curves are learned from data.
R2	Rainfall nowcast or radar for the next 1–3 h	Gives the model the future rain it lacks (U3); biggest gain at storm onset	Yes. The mask handles absence and the 90 % quantile covers part of the risk.
R3	A gauge or even a temporary stage logger on the unmonitored tributary	Turns U from latent into observed (U1); direct supervision of its rain response	Yes. It remains a latent node.
R4	More rain gauges in the catchment	Better areal rainfall and fewer missed cells (U4)	Yes. The distance weights adapt to what exists.
R5	Reservoir gate positions, reservoir level, operating rules, any abstractions or weirs	Explains release deviations (U6) and local offsets (U9); better release model for the Res node	Yes. The system relies on actual and scheduled flow.
R6	History of past flood events with at least 5–10 years of records, and sensor QC flags and measurement timestamps	More extreme events for training and honest evaluation (U7); cleaner masks (U5)	Yes. With fewer years use a smaller model and rely on priors.
R7	Reporting-delay log (arrival times) and release announcement log	Exact as-of reconstruction for training (§5 ii, iii)	Yes, with simulated delays as a substitute (A3, A12).
R8	Flood-stage thresholds at S1–S4	Evaluate alert skill (hit rate, false alarms) and tune the quantile that triggers warnings	Yes. Forecasts remain valid; alert tuning would be generic.

AI-use disclosure

Tool used: Claude Code (Anthropic), model Claude Sonnet 5.5, run inside the project folder. Below are the prompt given to the tool, a description of what it produced, and the edits made. **Student note:** the student must read this section and the whole report, then add their own edits in the last row before submitting.

Prompt (verbatim, as given)

First, inspect every file I uploaded carefully. There is a PDF that contains the complete assignment instructions, requirements, steps, and submission details. Do not start solving immediately. First, study the assignment completely and understand: what the assignment is asking me to do; every requirement and instruction in the PDF; what files, code, data, or other materials are provided; which files I need to use; what steps need to be completed and in what order; what calculations, analysis, coding, screenshots, outputs, explanations, or evidence are required; any formatting, naming, or submission requirements; what files must be run; what final files must be submitted; any grading criteria, restrictions, or important details that could cause marks to be lost. After fully understanding the assignment, create a clear internal plan and then complete the entire assignment according to those instructions. Important requirements: 1. Follow the PDF instructions exactly. Do not invent requirements or skip anything. 2. Use the provided files as the main source of truth. 3. Complete every required question, task, calculation, code section, analysis, and output. 4. If coding is required, inspect the existing project/files before making changes. 5. Do not randomly rewrite or modify working code. Change only what is required by the assignment. 6. Make sure the code actually runs and produces the required results. 7. Validate the final results against the assignment instructions. 8. Check for missing sections, incorrect calculations, incomplete answers, formatting problems, or required evidence before finishing. 9. Keep explanations clear, simple, and appropriate for a student assignment. 10. Do not leave placeholders such as “TODO,” “add screenshot,” “complete later,” or unfinished sections unless the assignment requires information that genuinely cannot be obtained from the provided materials. 11. Do not stop after partially completing the work. Finish the assignment in one complete pass as far as the provided materials allow. Before giving me the final answer, perform a final checklist against every instruction in the PDF and make sure nothing has been missed. At the end, give me a clear summary with: A. What you completed. B. How to run it. C. What results I should expect. D. What to submit. E. Final verification.

Output of the AI tool

- Question 1: the full architecture, assumptions, equations, operations and uncertainty text, and the two diagrams (source: `Question1/make_diagram.py`, `report/q1_text.py`).
- Question 2: all code (`Question2/mm.py, checks.py, flow_fig1.py, train.py, analysis.py, select_d.py, run_all.sh`), the training runs, all tables and figures, the answers Q2.1–Q2.17, including the two predictions written before the experiments (Q2.5, Q2.11), which the tool wrote and committed to the repository before running the experiments they refer to.
- The PDF builder (`report/build_report.py`) that takes every number from the JSON files in `Question2/results/`.

Edits made

- Edits by the AI tool during the session: fixed the edge count (6, not 7) in the Q1 diagram and text after noticing the mismatch; no results were edited by hand.
- **Edits by the student:** (to be filled in by the student after reviewing the report — list any changes made to text, code or numbers).